

Department of Computer Science & Engineering

Jashore University of Science and Technology
2nd year 1st Semester, B.Sc. Engineering Final Examination, 2022 Session (2019-20)

Course Title: Object Oriented Programming
Time: 3 Hours

Course code: CSE 2103
Full marks: 72.0

(Answer any six set of questions from the followings)

1. a. Distinguish between the following terms and provide suitable examples for each:

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- i) Objects and Classes ii) Data abstraction and data encapsulation
iii) Inheritance and polymorphism iv) Dynamic binding and message passing

b. What will be output of the below code? Write answer with explanation.

4

i) <pre>public class A { public static void main (String[] args) { if (5) break; } }</pre>	ii) <pre>public class A { public static void main (String[] args) { System.out.println('j' + 'a' + 'v' + 'a'); } }</pre>
iii) <pre>public class A { public static void main (String[] args) { int \$_ = 5; } }</pre>	iv) <pre>public class Demo{ public static void main (String[] arr){ } public static void main (String arr){ } }</pre>

2. a. Differentiate between Object Oriented Programming and Procedural Programming.

b. When do we use a protected access modifier? How can we prevent a class from being inherited by other classes?

c. What is the output of the following program? Provide reason(s) for your answer.

<pre>public class Vehicle { int id = 120; public int getId() { return id; } }</pre>	<pre>public class Car extends Vehicle { int id = 100; public int getId() { return(id - 20); } public static void main(String[] args) { Vehicle vc = new Car(); Car car = (Car) vc; System.out.print(vc.id + ", " + vc.getId() + ", "); System.out.print(car.id + ", " + car.getId()); } }</pre>
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3. a. What are the benefits of Encapsulation in OOP? Do you think encapsulation is properly ensured in the

following class? If not, please modify the code and ensure it.

```
public class Person {
    public int age;
    public String name;
    public String address;
}
```

- b. Explain the Is-a Relationships and Has-a Relationships in OOP. Let's say a system contains the following objects. Can you identify the relationship among these objects?

1. Bike	2. Plusar	3. Engine
4. Building	5. Window	6. Door
7. Student	8. Teacher	9. Person

- A. a. What is the signature of a method?

b. In the below Class X, is 'method' properly overloaded?

```
class X
{
    int method(int i, int d)
    {
        return i+d;
    }

    static int method(int i, double d)
    {
        return (int)(i+d);
    }

    double method(double i, int d)
    {
        return i+d;
    }

    static double method(double i, double d)
    {
        return i+d;
    }
}
```

c. What will be the output of the following program?

```
class X
{
    void method(int a)
    {
        System.out.println("ONE");
    }

    void method(double d)
    {
        System.out.println("TWO");
    }
}

class Y extends X
{
    @Override
    void method(double d)
    {
        System.out.println("THREE");
    }
}

public class MainClass
{
    public static void main(String[] args)
    {
        new Y().method(100);
    }
}
```

d. What are the differences between abstract class and interface?

12

- b. a. CSE, JUST wants to develop a **Canteen Management System**. Identify and implement possible classes of this system. Use inheritance if there are similar properties among the classes. You can use abstract class, interface and other oop components if necessary.

7

- ✓ 6. a. Write a program using any language that implements an abstract class Animal that has a Name property of type text and three methods SetName (string name), GetName and Eat. The Eat method will be an abstract method of type void.

You will also need to create a Dog class that implements the above Animal class and the Eat method that says the dog is Eating.

To test the program ask the user for a dog name and create a new Dog type object from the Main of the program, give the Dog object a name, and then execute the GetName and Eat methods.

5

- b. What are the differences between association, aggregation and composition?

5

7. a. The Collections class of Java Collection Framework has several methods (e.g. sort(), min(), max()) that require natural ordering of the classes to be handled. Explain the two possible conditions that must be satisfied on the classes to be able to use the sort() method to arrange the list of objects these classes.

- b. Explain why you would choose a LinkedHashSet (or LinkedHashMap) over a HashSet (or HashMap, respectively).

3

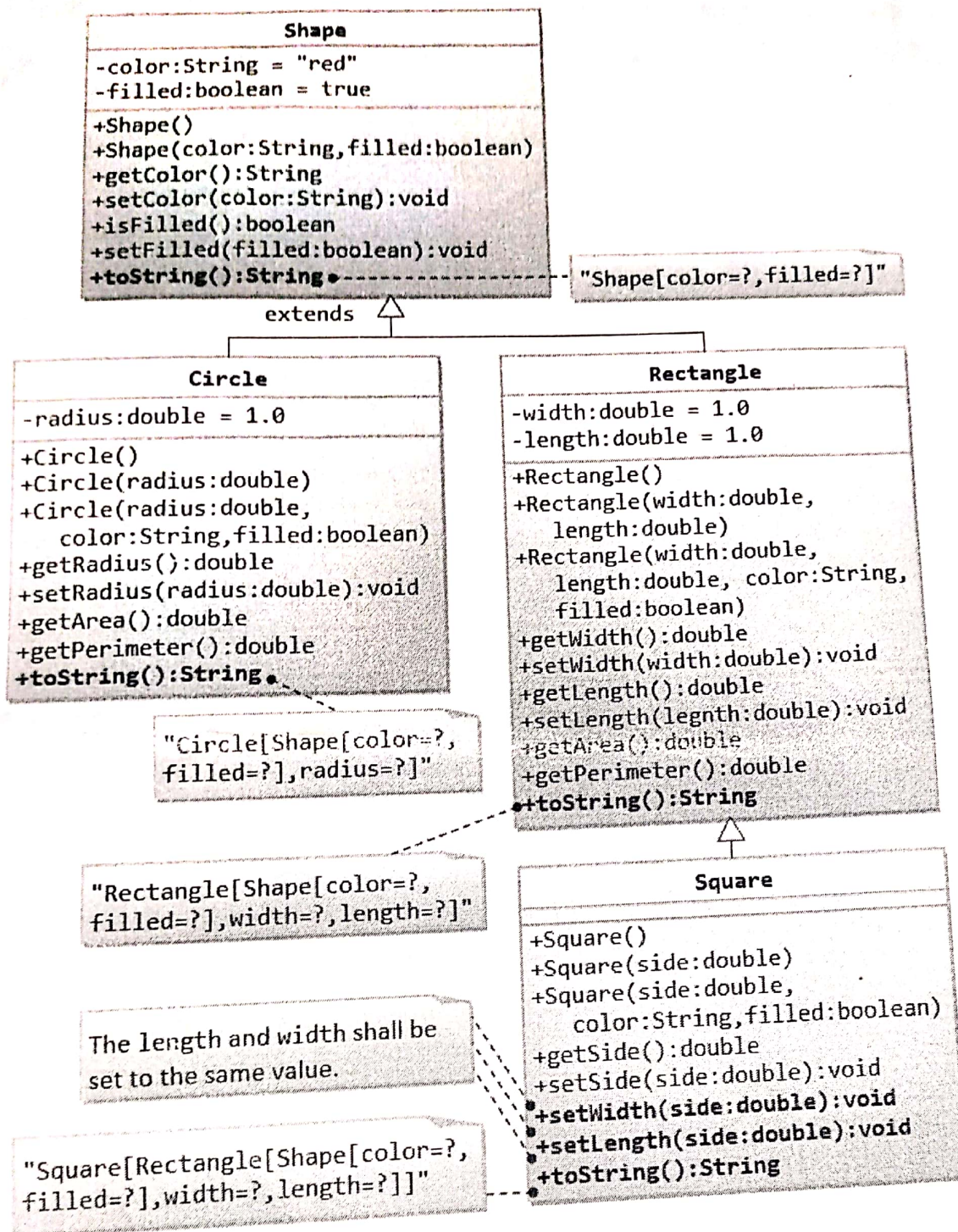
- c. Distinguish between

i) InputStream and Reader classes

ii) OutputStream and Writer classes

8. a. Write a superclass called Shape (as shown in the class diagram), which contains:

1. Two instance variables color (String) and filled (boolean).
2. Two constructors: a no-arg (no-argument) constructor that initializes the color to "green" and filled to true, and a constructor that initializes the color and fills to the given values.
3. Getter and setter for all the instance variables. By convention, the getter for a boolean variable xxx is called isXXX() (instead of getXxx() for all the other types).
4. A toString() method that returns "A Shape with color of xxx and filled/Not filled".



Department of Computer Science & Engineering

Jashore University of Science and Technology
2nd year 1st Semester, B.Sc. Engineering Final Examination, 2022

30

Course Title: Business Studies for Engineers
Time: 3 Hours

Course code: BUS 2101
Full marks: 72.0

(Answer any six set of questions from the followings)

1. a) Define administration. How administration differs from management? Describe. 1+4
b) What do you mean by organization? Briefly explain formal and informal organizational structure. 1+6
2. a) Differentiate between authority and responsibility. 4
b) What was Taylorism, and how did it give rise to Scientific Management Theory? 4
c) Describe about the major contributors to scientific management theory. 4
3. a) Suppose you have a private company to sell hardware products associated with vehicle. Your company is not making enough profit and you want to redesign a new structure for your organization. How you will do that? 6
b) What is span of control? Write the advantages and disadvantages of the narrow span of control and wide span of control. 1+5
4. a) Explain the meaning of economic laws according to Marshall and Robin. 3
b) Consider the nature of economic laws as behaviorist and Hypothetical and describe it. 4
c) Define engineering economics. Write down the characteristics of engineering economics. 1+4
5. a) Define management accounting. Differentiate between cost accounting and management accounting. 1+3
b) Explain the concept of management accounting. 4
c) Briefly describe the importance of management accounting in business sector. 4
6. a) The accounts of the Steelways Engineering Co. Ltd. show for 2010:

Materials used	\$1,80,000
Manual and machine labour wages directly chargeable	\$1,60,000
Works overhead expenditure	\$40,000
Embellishment and general expenses	\$19,000

- (i) Show the works cost and total cost, the percentage that the works overhead cost bears to the manual and machine labour wages and the percentage that the establishment and general expenses bear to the works cost. 4
- (ii) What price should the company quote to manufacture a machine which, it is estimated will require an expenditure of \$8,000 on materials and \$6,000 on wages so that it will yield a profit of 25% on the total cost or 20% on selling price. 3
- b) The following data relate to the manufacture of a standard product during the four-week period to June 30th, 2011: 5

Particulars	\$
Raw materials consumed	4,000
Wages	6,000
Machine hours worked	1,000
Machine hour rate	50 cents
Office overhead	20% on works cost
Selling overhead	6 cents per unit
Units produced	20,000
Units sold	18,000 @ Re. per unit

Prepare a cost sheet showing the cost per unit and profit for the period.

7. a) How important is motivation in the sense of organizational behavior? How motivation affects the performance of an employee in an organization? Explain. 6
- b) Suppose you have a Fast Track Courier service company and you are analyzing the cost factors and risk factors for your company. Which factors should you consider for this analysis? Describe 6
8. a) If you had to decide whether to continue making a component part or to begin buying the part from an overseas supplier, what quantitative and qualitative factors would influence your decision? 3
- b) Why do companies prepare budgets? 3
- c) How does managerial accounting differ from financial accounting? 2
- d) The administrator of Azalea Hills Hospital would like a cost formula linking the administrative costs involved in admitting patients to the number of patients admitted during a month. The Admitting Department's costs and the number of patients admitted during the immediately preceding eight months are given in the following table: 4

Month	Number of Patients Admitted	Admitting Department Costs
May	1,800	\$14,700
June	1,900	\$15,200
July	1,700	\$13,700
August	1,600	\$14,000
September	1,500	\$14,300
October	1,300	\$13,100
November	1,100	\$12,800
December	1,500	\$14,600

Required:

- i) Use the high-low method to estimate the fixed and variable components of admitting costs.
- Express the fixed and variable components of admitting costs as a cost formula in the form $Y = a + bX$.

50+



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B.Sc. Engineering Final Examination-2022

2nd Year 1st Semester, Session: 2019-2020

B.Sc. Engineering

Course Code: CSE 2107

Time: 3.00 hours

Course Title: Numerical Analysis

Marks: 72

[N.B. Answer any 6 set questions from following 8 set questions. The figures in the margin indicate full marks.]

1. a) Describe the Bisection method with proper figures. 04

b) Find the positive root of the equation $2x = 3 + \cos x$, by the bisection method, correct to the three decimal places. 04

c) Find a root of the equation $x^3 + x^2 - 1 = 0$ by using false position method, correct up to four decimal places. 04

2. a) Using Euler method, find an approximate value of y where $0 \leq x \leq 0.5$ and $h = 0.1$, given that $\frac{dy}{dx} = x^2 + y^2$ and $y(0) = 1$ 04

b) Find a real root of the following equations, by the Newton-Raphson method: 04
 $x^3 + 3x + 10 = 0$

c) Find a real root of the following equations, by the Secant method: $x^3 + 3x + 10 = 0$ 04

3. a) Given the following table. Calculate the $f(35)$ using forward difference 06

x	21	25	29	33	37
f(x)	18.47	17.81	17.10	16.34	15.51

b) Write a program for calculating the forward difference table of the above table. 06

4. a) Solve the following system of equations using the Gauss Elimination method: 06
 $2x_1 + 2x_2 + x_3 = 6$

$$4x_1 + 2x_2 + 3x_3 = 4$$

$$x_1 + x_2 + x_3 = 0$$

b) Write a program for above question. 06

5. a) What are the main objectives of numerical analysis? 02

b) Calculate the numerical integration using Simpson's rule of $f(x) = 3x^2 + 5x + 2$. Here, n , lower and upper limits are 8, -2 and 5, respectively. 06

c) Find the absolute error in $w = xy + z$ if $x = 3.35$, $y = 5.74$ and $z = 2.45$ 04

6. a) What are the main limitations of the bisection method?

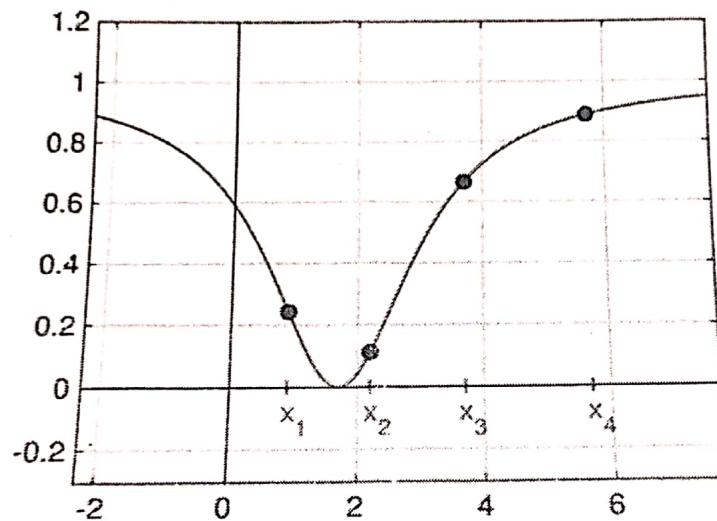
02

b) Compute the Taylor polynomial approximation to $\sin(x)$ expanded about the point

06

$x = \pi/8$.

c) For the function $f(x)$ plotted below, use each of the four points x_1 , x_2 , x_3 and x_4 as the starting guess for Newton's method. For which point(s) do you expect the iteration to converge to the root shown? Also provide reasons for your answer.



7. a) Find the root of $f(x) = 3x + \sin(x) - ex$ using the Muller method.

06

b) Write a program for finding the root of the above equation.

06

8. a) Find the root of $f(x) = 3x + \sin(x) - ex$ using Hallie's method.

06

b) Write a program for finding the root of the above equation.

06



Course Code: CSE 2101

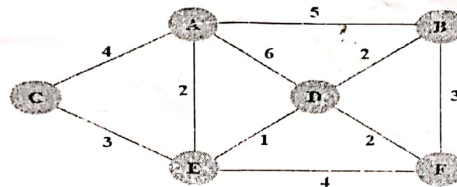
Course Title: Introductions to Algorithms

Time: 3.00 hours

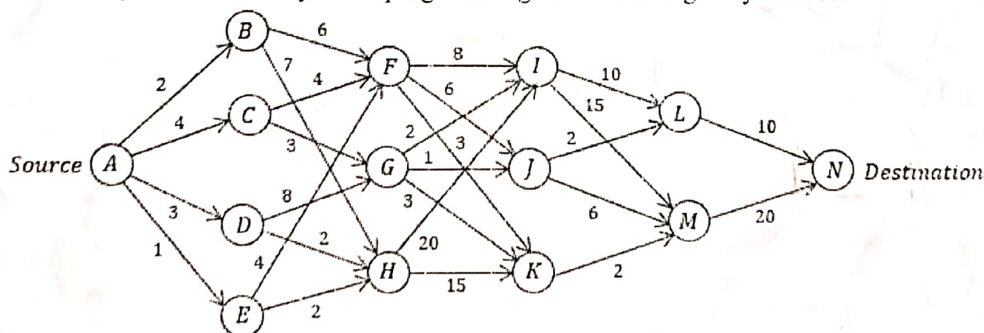
Marks: 72

[N.B. Answer any 6 questions from following 8 questions. The figures in the margin indicate full marks.]

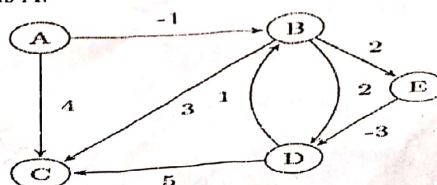
- 2/6 a) How can you measure the efficiency of an algorithm? Describe Euclid's algorithm. 05
b) Define $\Omega(n)$ and $\Theta(n)$. Show that following equalities are correct 04
i. $n^{4.5} + n^e \log n = \theta(n^{4.5})$ ii. $n^3 + 10^6 n^2 = \theta(n^3)$
c) Define binary search. From the array find the positions of -6, and 125 using binary search. 03
Show the steps. -15, -6, 0, 7, 9, 23, 54, 82, 110, 120
- 12/2 a) Write how divide and conquer method works. Write an algorithm to find the maximum and 07
minimum from a list of integers using divide and conquer method.
b) Show the steps used to sort the following data using merge sort technique in ascending order: 310, 05
285, 179, 652, 351, 423, 861, 254, 450 and 520.
- 11/3 a) Define Greedy Algorithms and Write the elements of the greedy strategy. 03
b) What is an optimal Huffman code for the following set of frequencies? Show fixed and variable 04
length bits/character.
a:1, b:1, c:2, d:3, e:5, f:8, g:13, h:21
c) What is MST? Find minimum spanning tree by applying prim's algorithms on following weighted 05
graph where the root vertex is A



- 4/ a) Consider the following set S of activities: 04
- | | | | | | | | | | | | |
|-------|---|---|----|---|---|---|---|----|----|----|----|
| S_i | 0 | 1 | 2 | 3 | 3 | 5 | 5 | 6 | 8 | 8 | 12 |
| f_i | 6 | 4 | 14 | 5 | 9 | 7 | 9 | 10 | 11 | 12 | 16 |
- Now sketch the operation of recursive activity selector algorithm and write the greedy activity selector algorithm.
- b) From the figure show that dynamic programming is better than greedy method. 06

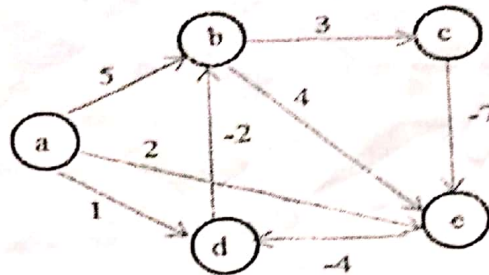


- c) Define Dynamic Programming and write the sequence of steps to develop a dynamic programming 02
algorithm.
- 5/ a) Find out the Longest Common subsequences for the following sequences: 04
X:BCDAACD
Y:ACDBAC
b) Show the executions of Bellman-Ford algorithms for the following weighted directed graph 05
where the source vertex is A.



- c) What is all pair shortest path problem? Show an example. 03

- ✓ a) Find the sequence of matrices $D^{(k)}$ computed by the Floyd-Warshall algorithm for the following weighted directed graph. 05



- b) Write the differences between BFS and DFS. 03
 c) Describe the recursive top down and bottom up rod cutting algorithm. 04
7. a) For a binary tree traversal, if the InOrder and PreOrder traversal show the following node sequences, then find the node sequence for the PostOrder method after constructing the tree. 06
 InOrder: 4, 2, 5, 1, 3, 6
 PreOrder: 1, 2, 4, 5, 3, 6
- b) From the multistage graph of fig. 7(a) show that dynamic programming is better than greedy method. 06

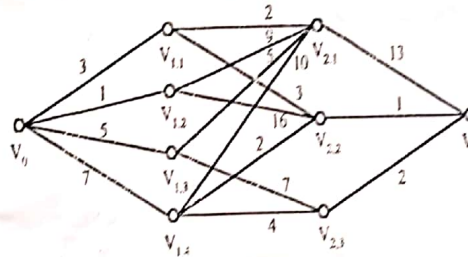
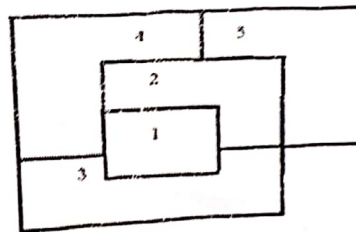


Fig. 7(a)

8. a) What is Backtracking? Describe backtracking solution of the 4-queen problem by drawing the chess board and state space tree. 06
 b) What is planner graph? Represent the following map into a planner graph. How many colors are needed to color it? 03



- c) "All NP-complete problems are NP-hard, but some NP-hard problems are not NP-complete"-Justify. 03